UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING
28 FEB 2022
EXAMINATION & REGISTRATION

UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

U. C. S. C. LIBRARY UNIVERSITY OF COLOMBO

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester I – 2021/2022

SCS1201 – Data Structures and Algorithms I

TWO (2) HOURS (For Part A & B)

058

PART (A)

To be completed by the candidate

Examination Index No: .....

Important Instructions to candidates:

1. The medium of instruction and question is **English**.
2. Write your answers in English
3. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
4. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
5. Write your index number on each and every page of the Question paper.
6. Part A has 03 questions in 14 pages.
7. Answer **ALL** questions.
8. **Part A** of the paper will total to **60 marks (20 marks for each question)** and **Part B** of the paper will total to **40** marks.
9. **This paper consists of two parts, Part A (Question No 1, Question No 2 and Question 3) and Part B (Question No 4 only) and submit both Parts separately.**
10. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed.**
11. **Non-Programmable** calculators are **allowed.**

For Examiner's use only	
Question No	**Marks**
1	
2	
3	
4	
Total	

Index No: ...

Part A

Question 1

(a) Consider the following algorithms:

- Tower of Hanoi
- Fibonacci number series
- Tree traversal

One student proposed the following conclusion after carefully considering the implementation of the above algorithms.

"The above algorithms can be implemented using only recursive approach"

As a skilled student of implementation of Data Structures and Algorithms, discuss the validity of the above conclusion.

[4 Marks]

(b) The quality of an algorithm depends on the following characteristics.

- Unambiguous.
- Input
- Output
- Finiteness
- Feasibility
- Independent

Discuss how one would incorporate the above characteristics when designing a quality algorithm?

[6 Marks]

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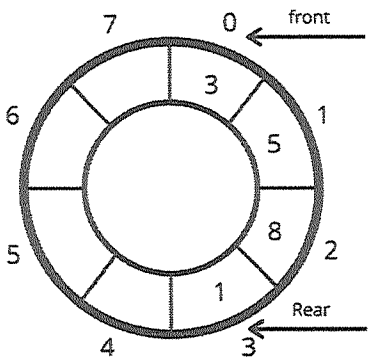
(c) Consider the following circular queue with initial values.



Initails values are :
Front=0
Rear=3
Count=4

(i) Write Front, Rear, and Count values after two (02) data elements are deleted from the Queue.

[2 Marks]

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Front =
Rear=
Count =

(ii) Write Values for Front, Rear, and Count after insertion of following data elements to the queue obtained from Part C(i) above.

Data elements are : 6,2,7,9.4

[2 Marks]

Front=
Rear=
Count=

(d) Consider that one processor is shared between 5 terminals (you may assume that the terminals do not have a processing power). Suppose the terminals submit their jobs to the processor according to the following terminal ID order.

i. A
ii. B
iii. C
iv. D
v. E

After submitting their jobs to the processor, the operating system will generate a priority value, and the priority values with terminal ID's are shown in the following table.

Terminal ID	A	B	C	D	E
Priority Value	8	12	3	5	22

(i) Construct a Max-Heap based on the initial submission order. You may apply the top-down fixing approach if there are any violations of the heap property.

[3 Marks]

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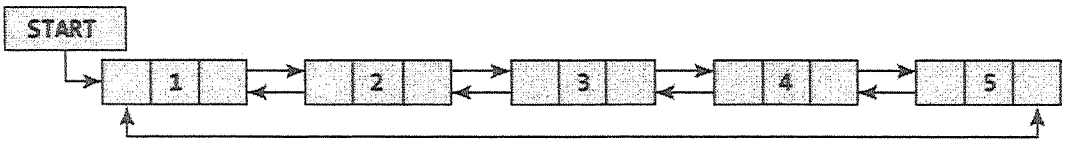
(ii) Illustrate how to sort the jobs in different terminals according to the increasing order of priority values without using an additional array.

[3 Marks]

Index No: ...

Question 2

(a)Consdier the following circualr doubly linked list.



(i) Write a Pseudocode or C code to find a given node from the above-linked list.

[3 Marks]

(i)

(ii) Write a Pesudocode or C code to delete the first Node (i.e.data element =1) form the above linked list.

[3 Marks]

(b) Using a suitable example(s), show the validity of the following statements.

Note: h is the height of the Binary tree

(i) The total number of leaf nodes in a complete binary tree is at least h+1 and at most 2^h

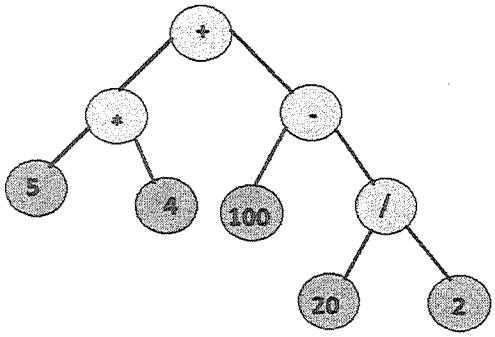
(ii) The total number of nodes in a complete binary tree is at least 2h+1 and at most $2^{h+1} - 1$

[2x2=4 Marks]

Continued.

Index No: ...

(c) Consider the following expression tree.



What is the order of nodes of the trees if traversed in the following sequence?

(i) Pre-order
(ii) In-order
(iii)Post-order

[4 Marks]

Index No: ...

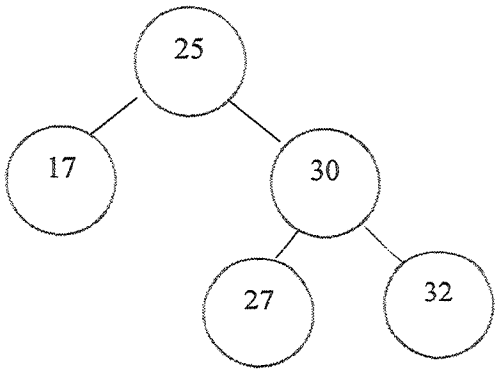
(i)
(ii)
(iii)

(d) The following algorithm is used to insert new nodes into Binary Search Tree.

```
Input: binary search tree T, node v, element e
Output:
Insert(T, v, e){
        if(T.isLeaf(v)){
                if(v.element()>=e)
                        Insert element e as v's left child
                else
                        Insert element e as v's right child
        } else {
        if(v.element()>=e)
                        Insert (T, T.leftChild(v), e)
                else
                        Insert (T, T.rightChild(v), e)
    }
}
```

Note: The content of the new node is denoted by e.

Illustrate how the above **pseudocode algorithm works** when inserting the given nodes to the existing tree.

(i) Existing tree	(ii) Existing Tree
	
New node to be inserted is 30	New node to be inserted is: 35

[3x2=6 Marks]

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Question 3

(a) Define **Self-Balancing Binary Search Trees.**

[3 Marks]

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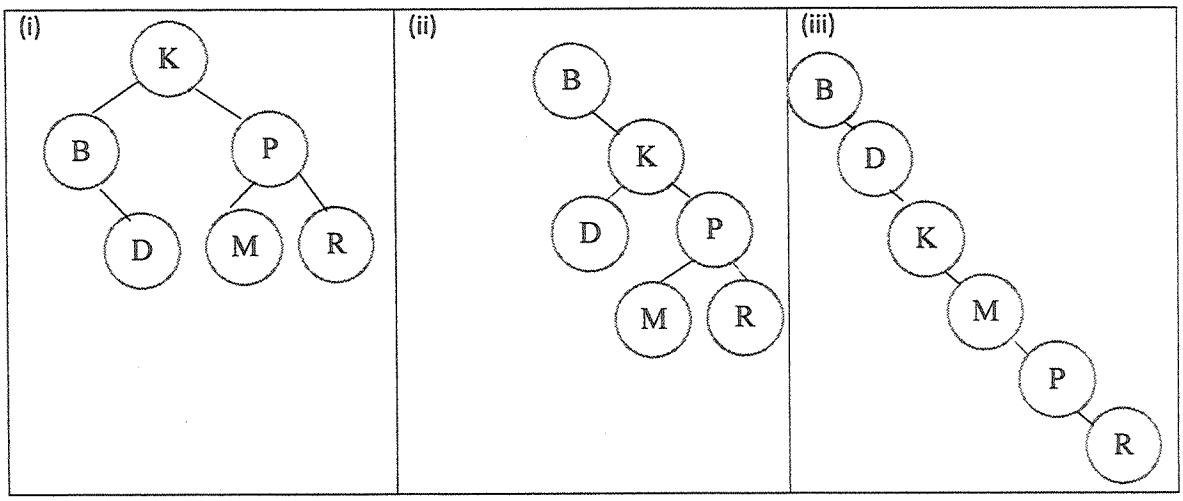
(b) Determine whether the following trees are balanced or not. If a tree is not balanced, discuss why it is not balanced.



(2x3=6 Marks)

(c) Day–Stout–Warren algorithm (DSW) algorithm for balancing a tree has two significant steps as follows:

Step 1: First, create a backbone or vine
Step 2: Transform the backbone/vine to a well-balanced tree.

Note: *For the two questions C(i) and C(ii) given below* you should provide all the intermediate steps when creating the backbone (part C(i)) and transforming it into a well-balanced tree (part C(ii)).

(i) Consider the following algorithm, which illustrates Step 1 of the DSW Algorithm.

```
CreateBackbone(root,n)
  Tmp=root
  While (Tmp! =null)
    If Tmp has a left child
       Rotate this child about the temp
       Set Tmp to the Child which Just Become parent
    Else Set Tmp to its right child
```

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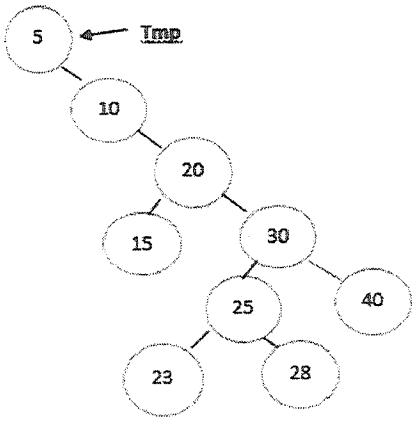
Transform the following binary search tree into a backbone/vine using the above algorithm.



[4 Marks]

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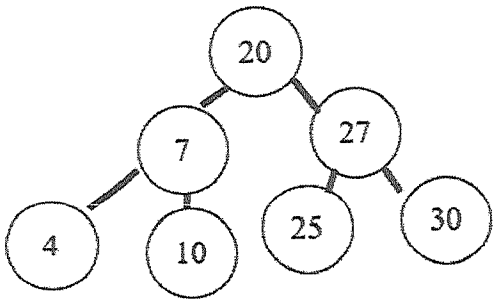
(ii)Using Suitable rotations, transform the backbone/vine created in Part (c) (i) above into a well-balanced tree.

[4 Marks]

(d) Consider the following AVL tree.



Insert the following nodes to the above tree.

23, 26, 21

It was noted that the AVL properties are violated after inserting the new nodes. Discuss how the AVL tree can be restored using suitable rotations.

[3 Marks]

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